Room Exchange — User Guide

Revit rooms to EASE 5 room models · BIM Acoustics AV Tools Suite
v3.0.0, November 2026

Room Exchange turns the rooms of a Revit model into an EASE 5 room model: a DWG of closed faces, one layer per surface material, that EASE imports as a closed volume. You tick the rooms, Room Exchange fills their air, finds every surface the air touches, names each face by the material on the room side, and writes the drawing. Every layer becomes a Selection Set in EASE, so materials are assigned per layer.

This guide covers the one thing that decides whether an export comes out right, the rooms you tick, and then the window, the checklist, the result and how to read it in EASE.

Room Exchange requires AutoCAD on the same machine. Room Exchange writes the model as a DXF and uses AutoCAD only to convert that DXF to the DWG that EASE 5 imports; nothing else is done in AutoCAD. The DXF export (Bose Modeler, EASE 4) does not use AutoCAD.

Room Exchange comes with the RoomExchange plan ($80/month or $800/year) and with AVTools Suite Pro ($99/month or $990/year). It also runs during the free 14-day trial, which starts when you first open System Designer and register. After buying either plan, enter your subscription email in **License Manager** (BIM Acoustics tab → Utilities) and click **Validate Now**; **Manage or Upgrade Subscription** there moves you from one plan to another. System Designer's Room Acoustics tab uses the same engine to find the surfaces for its RT60, one room and one level at a time; a space drawn as several stacked rooms is treated as one only when you export them together here.

1. The rooms must envelope the whole space

Room Exchange fills air. It only fills the air that a ticked Revit room claims. Where the room stops, the fill stops, and the edge of the room becomes a flat face on the `EASE-$BOUNDARY` layer, whether or not there is a wall, floor or ceiling there in the model.

So before you export, make sure that the ticked room or rooms, together, reach every surface of the acoustic space: the lowest floor, the deck or roof above, and every wall. If they do, every face in EASE is a real surface. If they do not, EASE shows a box where the room limit was, and the volume is wrong.

What the architect's rooms usually look like

Architects place rooms for area and scheduling, not for acoustics. In a theater that typically means one room on the house floor to a nominal height, one on the stage, one for the catwalk, one for the pit, each stopping at its own level. In section they do not meet: there is a void between the top of the house room and the catwalk room, and the raked seating may sit in no room at all.

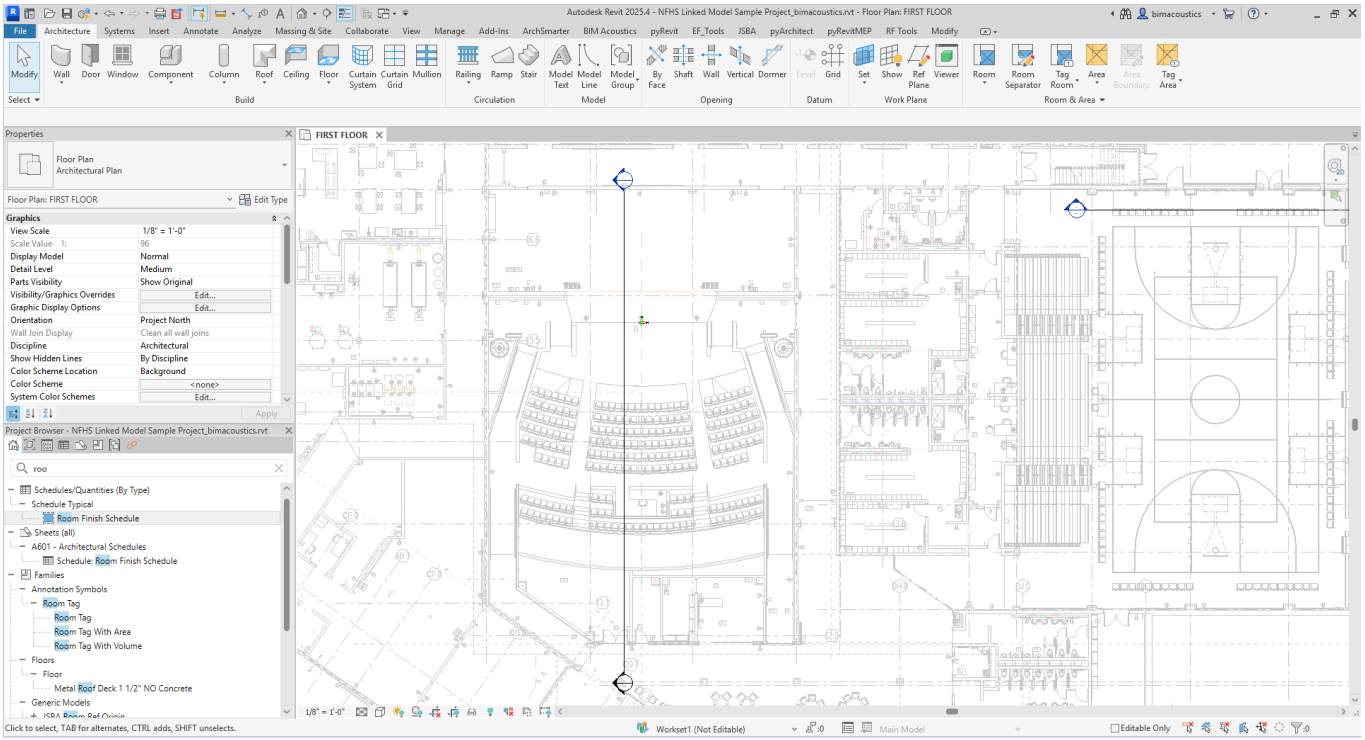


Figure 1. The theater in plan with the architect's rooms as delivered.

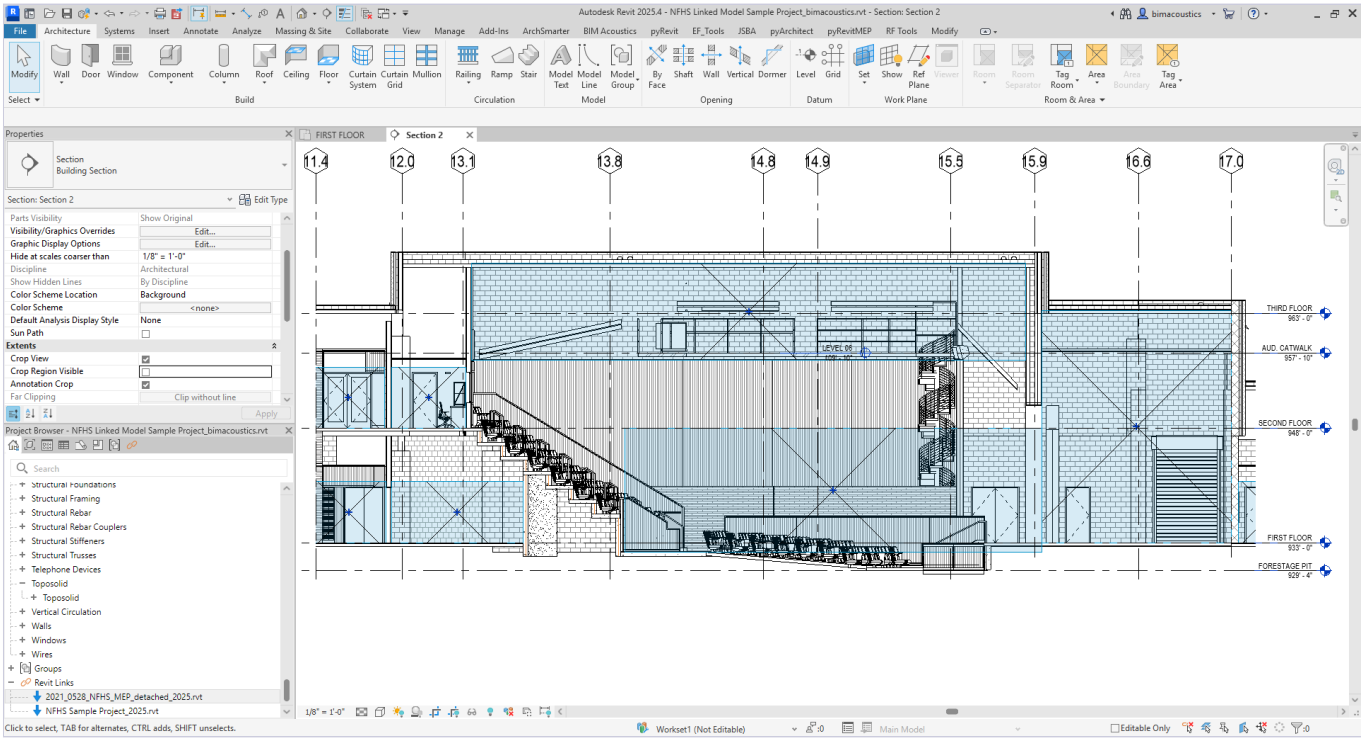


Figure 2. The same theater in section. The architect's rooms (light blue) stop at their own levels; the volume between them belongs to nobody.

Ticking those rooms and exporting gives you exactly what they claim: the fill reaches the top of each room and stops, the result dialog says the air reaches the limit toward the catwalk room over thousands of square

feet, and EASE shows a flat lid where the house room ended.

The fix: one room of your own that spans the space

Place a room of your own in the host model (your AV or acoustics model, not the architect's link) that covers the whole acoustic volume. One room is enough when the space is one volume; the architect's rooms can stay ticked alongside it, since rooms that touch are merged into one export.

1. **Pick the level and footprint.** Place the room on the level whose plan shows the full footprint. In the theater that was the second floor, where the house and the stage are one open plan. Let the room fill the house and the stage; room-bounding walls close it. If it leaks into a corridor or lobby, draw room separation lines across the opening.

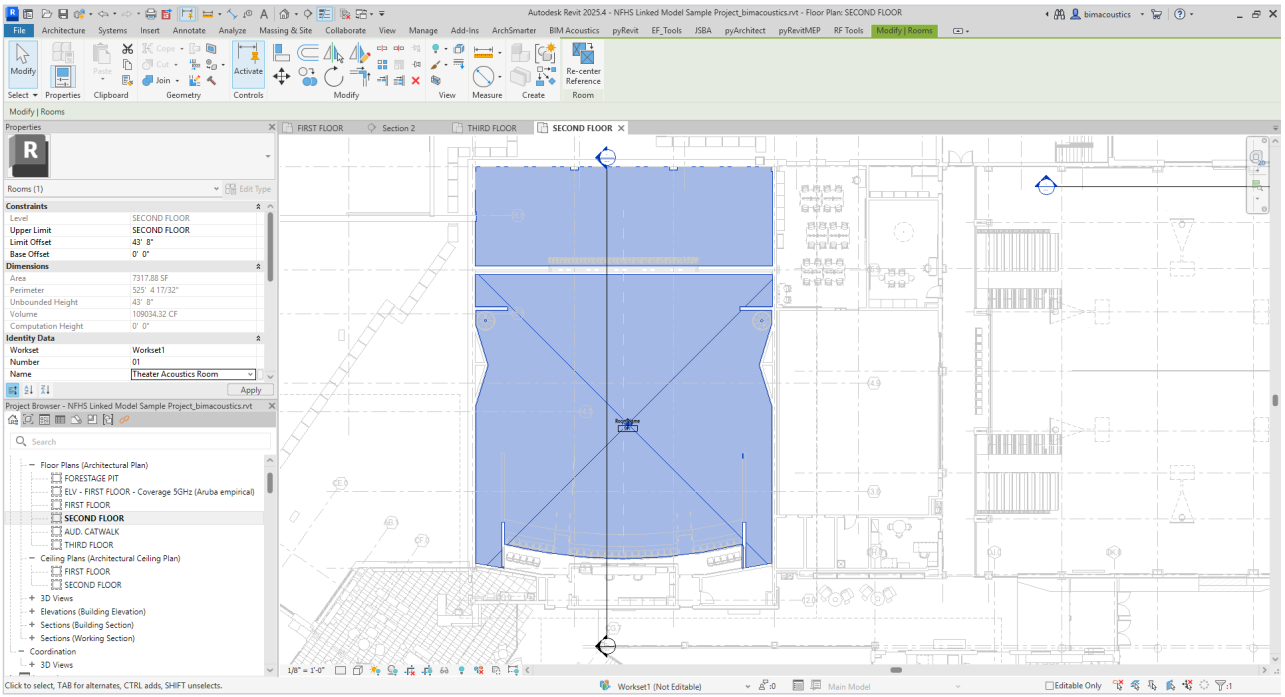


Figure 3. The acoustics room placed on the second floor, covering the house and the stage in one footprint.

2. **Set its height to the deck.** In the room's properties set Upper Limit and Limit Offset so the room reaches the underside of the roof deck. Check it in a section: the room's blue extent must touch the deck everywhere. A ceiling in the model will stop the fill where a ceiling exists; the room limit only has to be high enough.

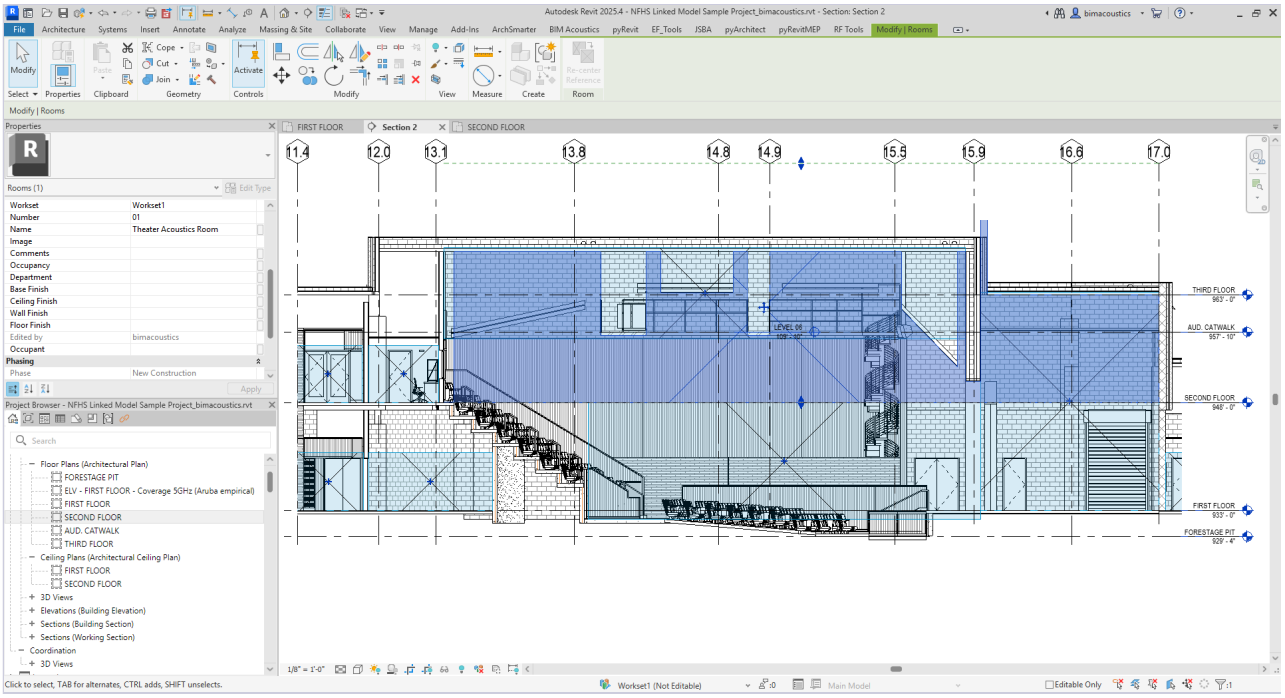


Figure 4. In section, the room's upper limit raised to the deck. It covers the upper house and the catwalk level, but not yet the lower house and the pit.

3. **Lower its base to the lowest floor.** Set the Base Offset (negative) so the room also reaches the lowest floor of the space, here the forestage pit. The raked seating, the pit and the stage floor must all lie inside the blue extent.

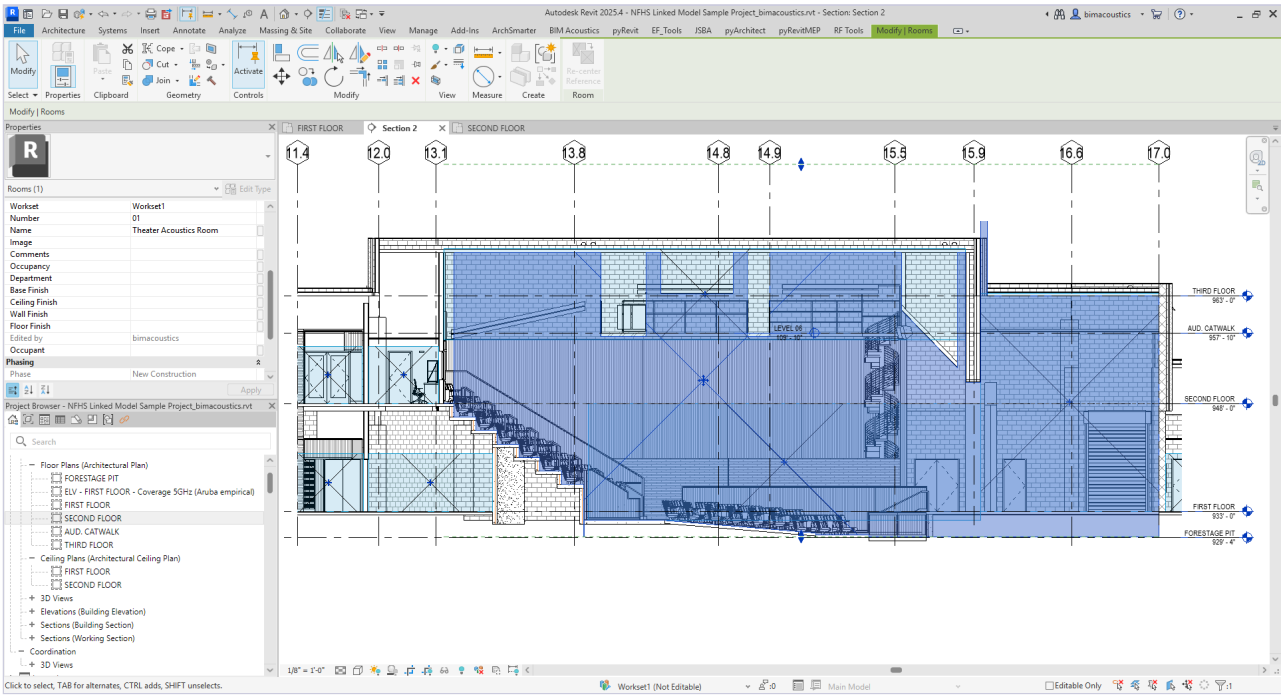


Figure 5. The same room with its base lowered: from the pit to the deck, every surface of the theater is inside the room.

4. **Name it so you can find it**, for example "Theater Acoustics Room", and give it a number that sorts to the top of the list. Put it on its own workset if other disciplines share the model.

A room that spans several levels is fine for Room Exchange; it does not need to be bounded at every level, only to reach every surface. For a balcony house, one tall room usually does; where a balcony slab divides the volume into two, use one room above and one below and tick both.

One footprint is not always enough: tick the architect's rooms as well

A Revit room has one plan footprint, taken at the level it was placed on. Anything that belongs to the space but sticks out of that footprint at another level is outside the room: a vomitory tunnel under the seating, a slot under a balcony, a pit that reaches further than the stage above it. The export of the theater with the placed room alone came out closed, but the south-west vomitory was cut off at the room's edge.

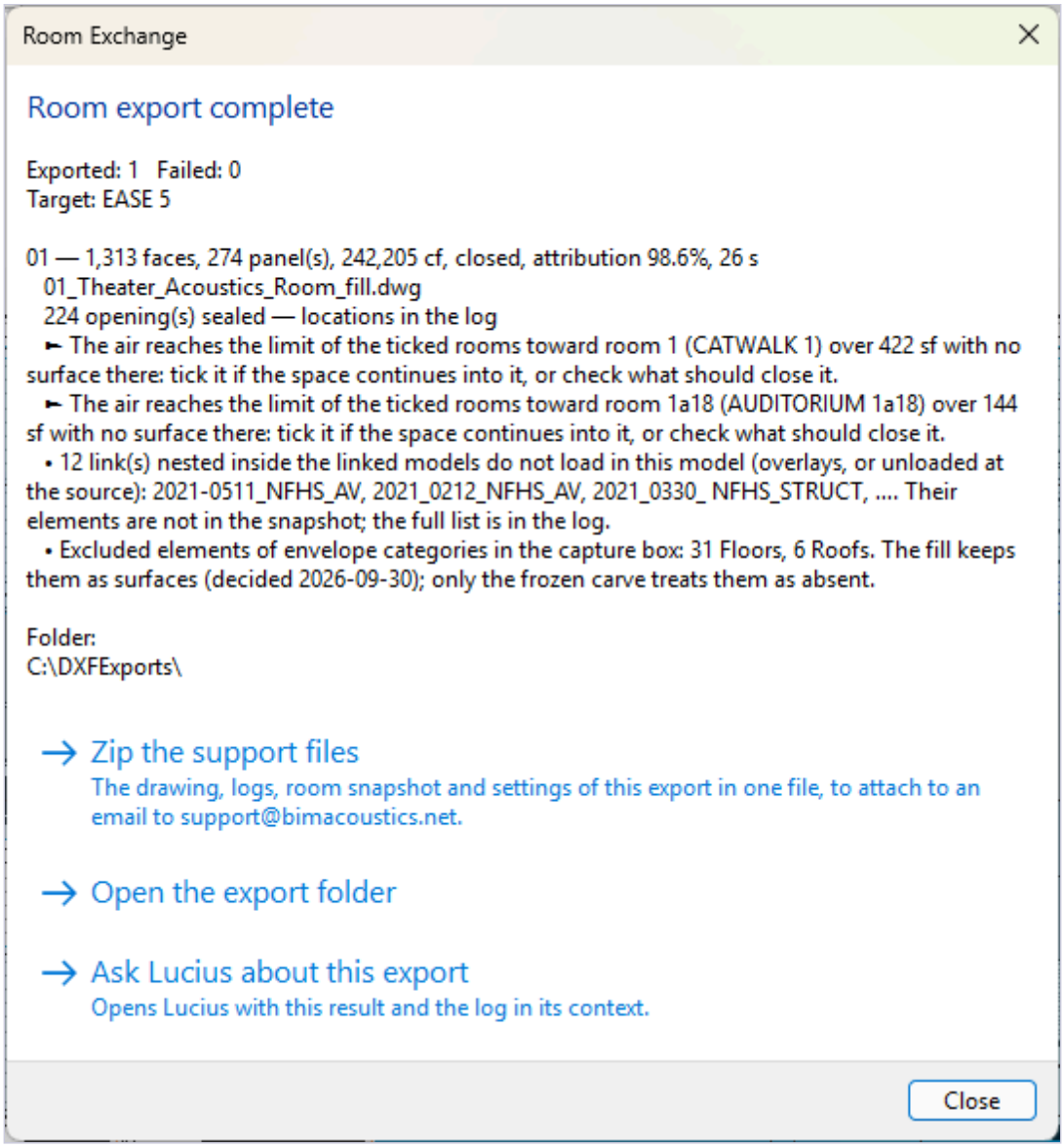


Figure 6. The placed room exported alone: closed, but the result dialog reports air reaching the limit toward the auditorium room over 144 sf, which is the vomitory.

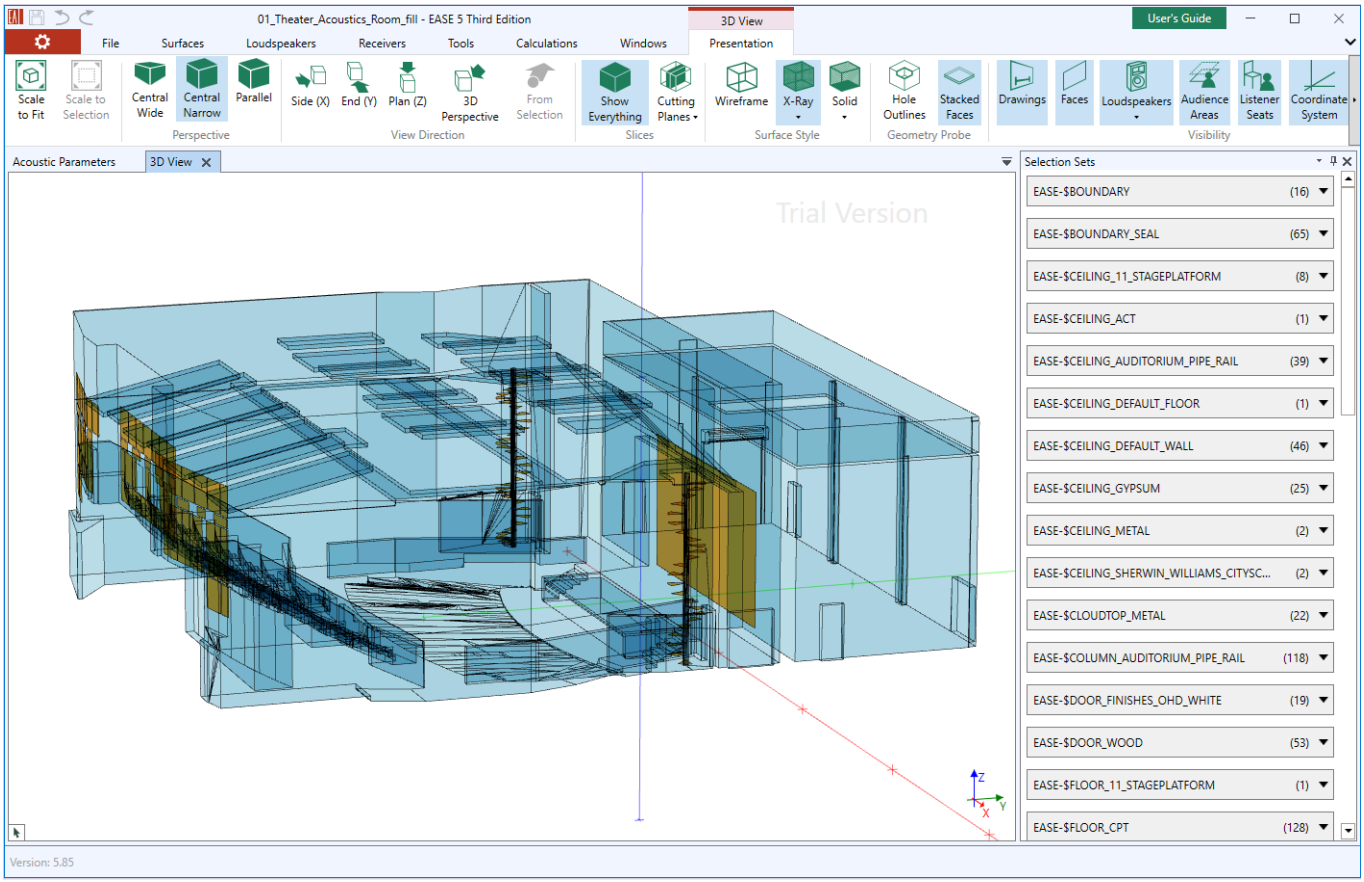


Figure 7. In EASE the same export: the vomitory tunnel ends in a flat boundary face where the placed room's footprint stopped.

So tick the architect's rooms that lie inside or against your room as well. Rooms that touch are merged into one volume, and each one contributes its own footprint: the auditorium room on the first floor brings the vomitory, the stage room brings the stage, the catwalk room the catwalk. Your room supplies the height; theirs supply the odd shapes at each level. Ticking them does no harm where they overlap, and it brings their names into the file name and the log.

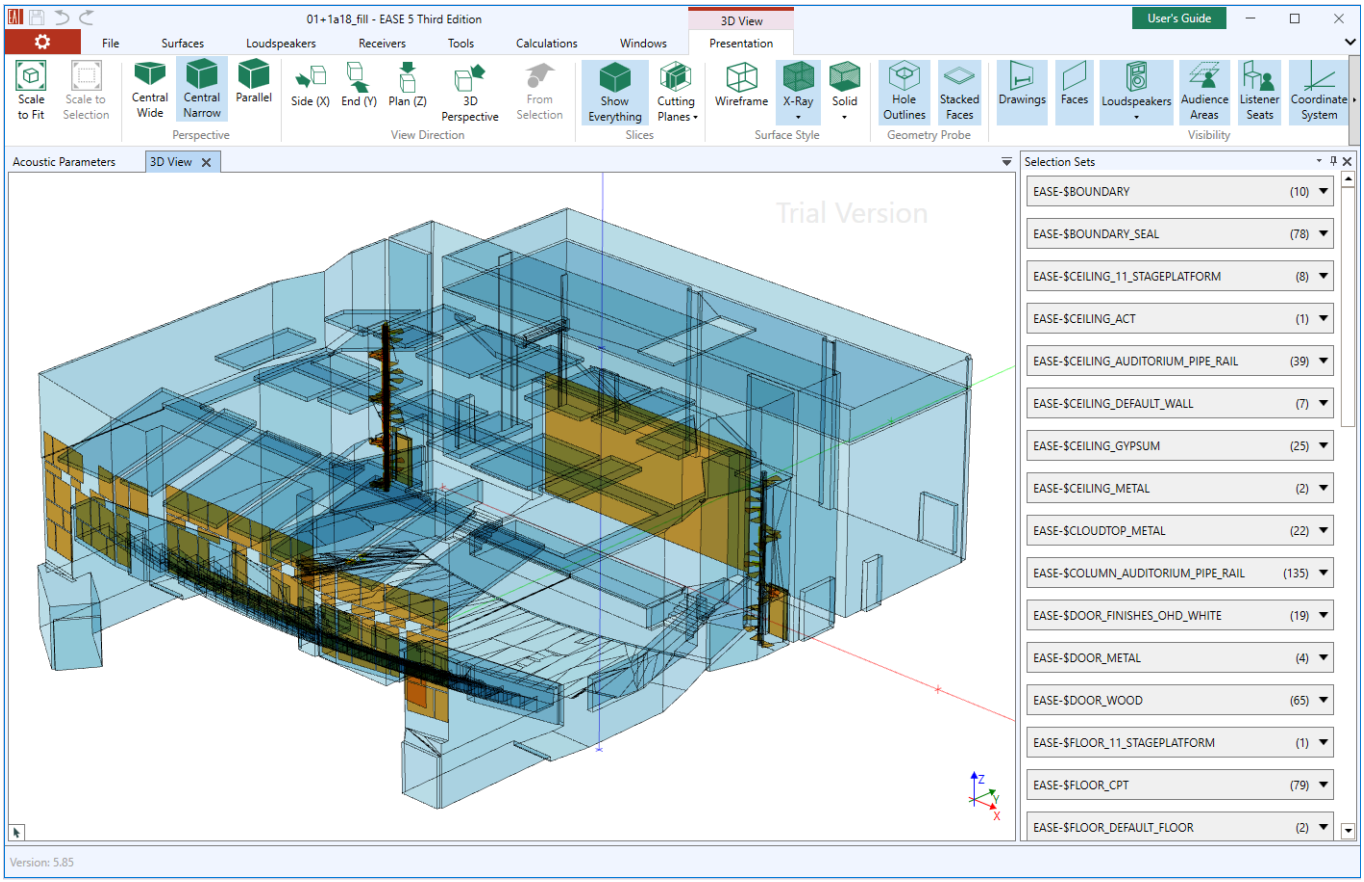


Figure 8. The placed room exported together with the auditorium room: the vomitory is a complete tunnel, and the volume is still closed.

The result dialog tells you when a footprint is missing: a line reading **the air reaches the limit of the ticked rooms toward room X over N sf** names the room to tick.

A room drawn round an opening in the floor, such as a stair or an atrium opening, has a hole in its footprint. When nothing stands in the hole, Room Exchange counts it as part of the room's air and closes it at floor level, so the opening does not come out as a box up to the roof. A hole that a column or a core stands in stays out of the room.

When no architect's room covers a part of the space either, place another room of your own there, on the level where that part has its footprint: a room in a vomitory, under a balcony, in a side box, in an orchestra pit wider than the stage. Complex spaces may take three or four rooms before every surface is inside one. There is no cost to more rooms; they merge into one volume when they touch, and the export is named after all of them unless you type a file name.

Export the rooms together

In Room Exchange tick your room and the architect's rooms it should take in; here the catwalk, the stage and the auditorium were ticked with it.

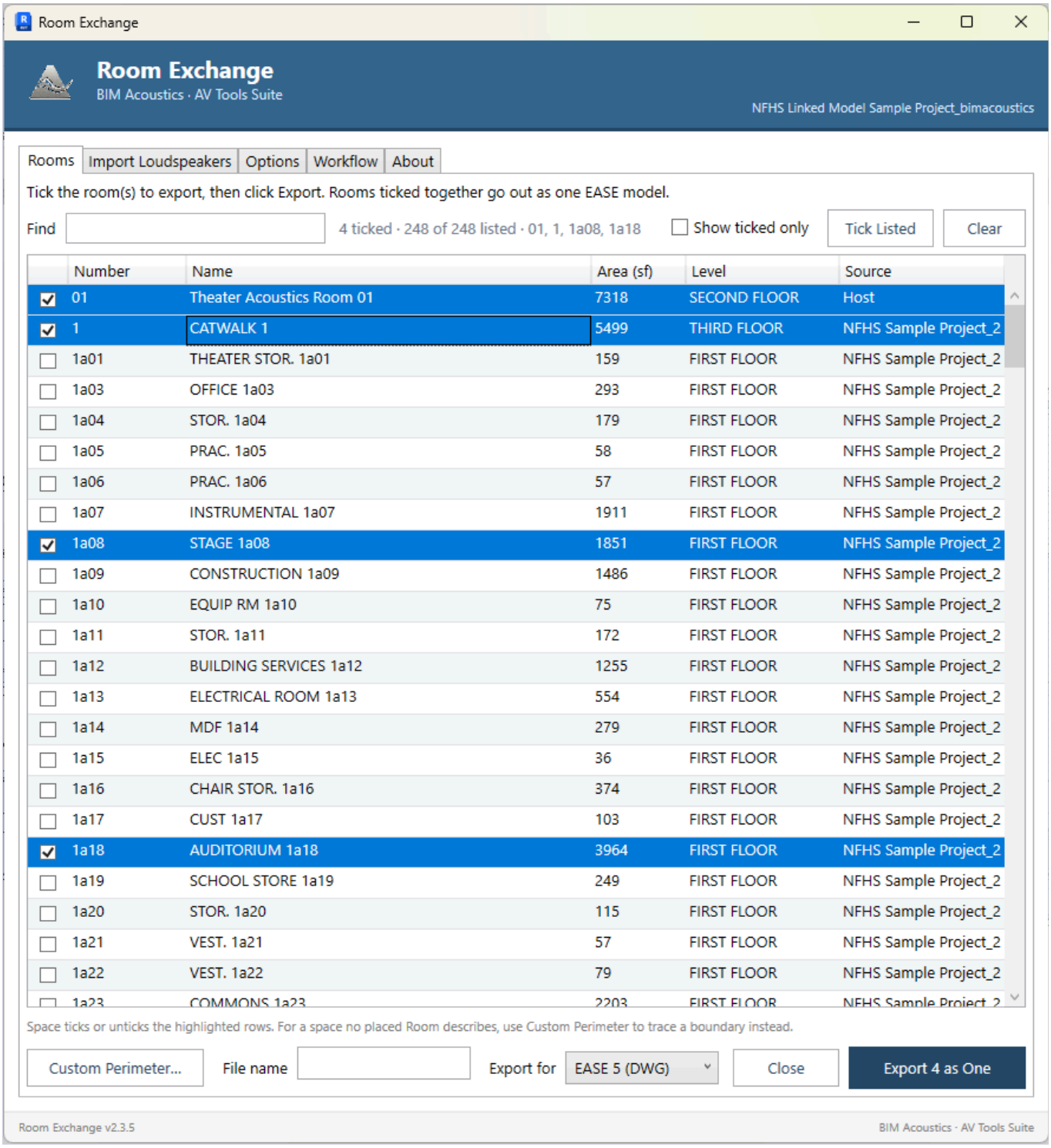


Figure 9. The Rooms tab with the acoustics room and the three architect's rooms ticked; the button reads Export 4 as One.

Rooms that touch are exported as one merged volume. Room Exchange asks once; answer Yes for a theater's house, stage and pit, or any set of rooms that are acoustically one space.

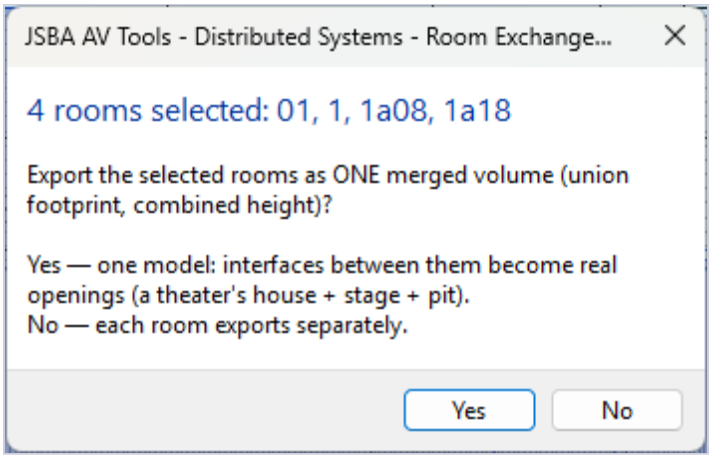


Figure 10. The merge prompt: one model, with the interfaces between the rooms as real openings.

The result of a room that envelopes the space

The checklist appears next (section 4). After it, the result dialog shows one block for the merged volume. Look for **closed** and a high attribution; the openings sealed are small gaps the fill closed.

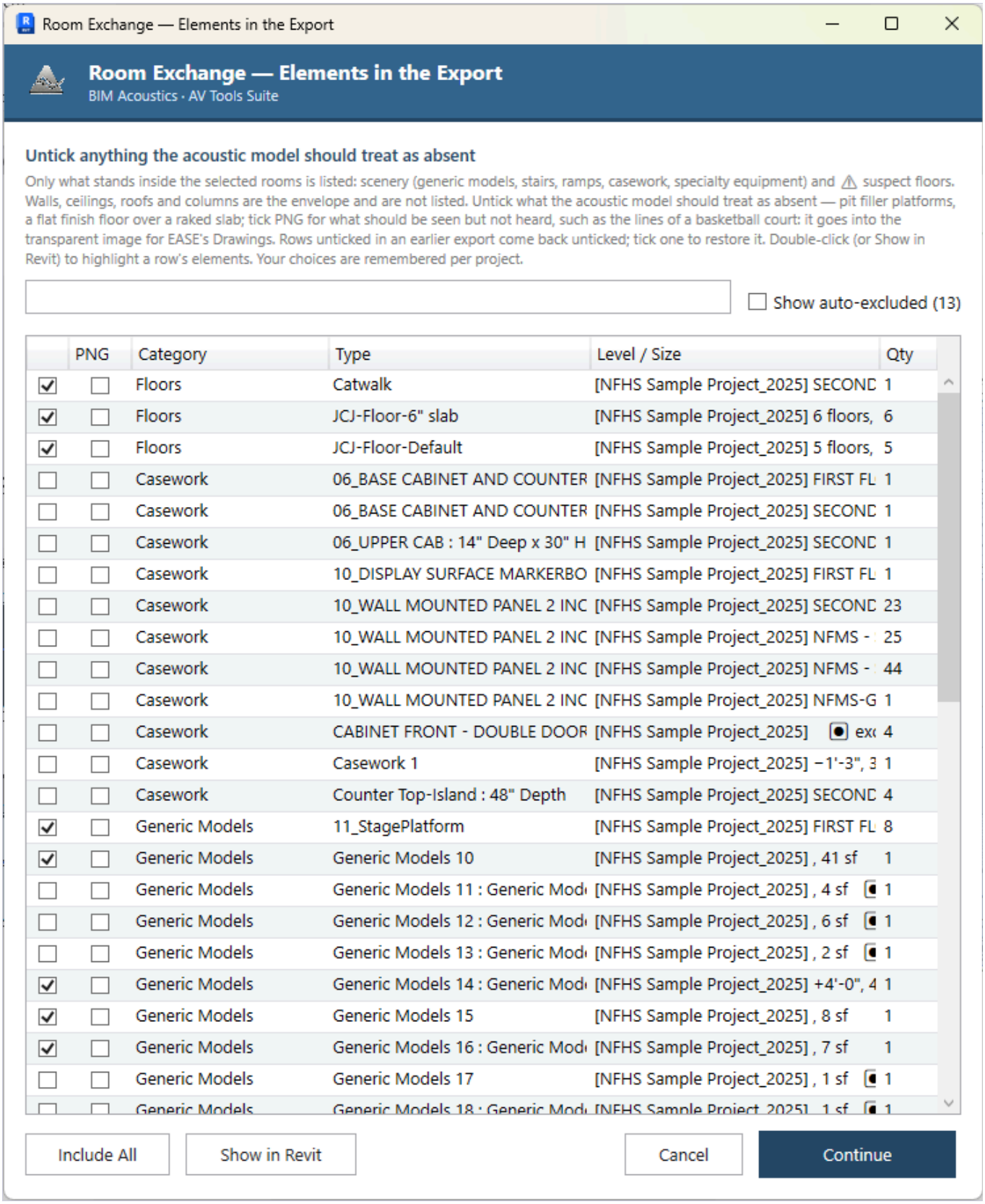


Figure 11. The checklist for the theater: the floors, stage platform and the generic models that are structure stay ticked; casework is left out.

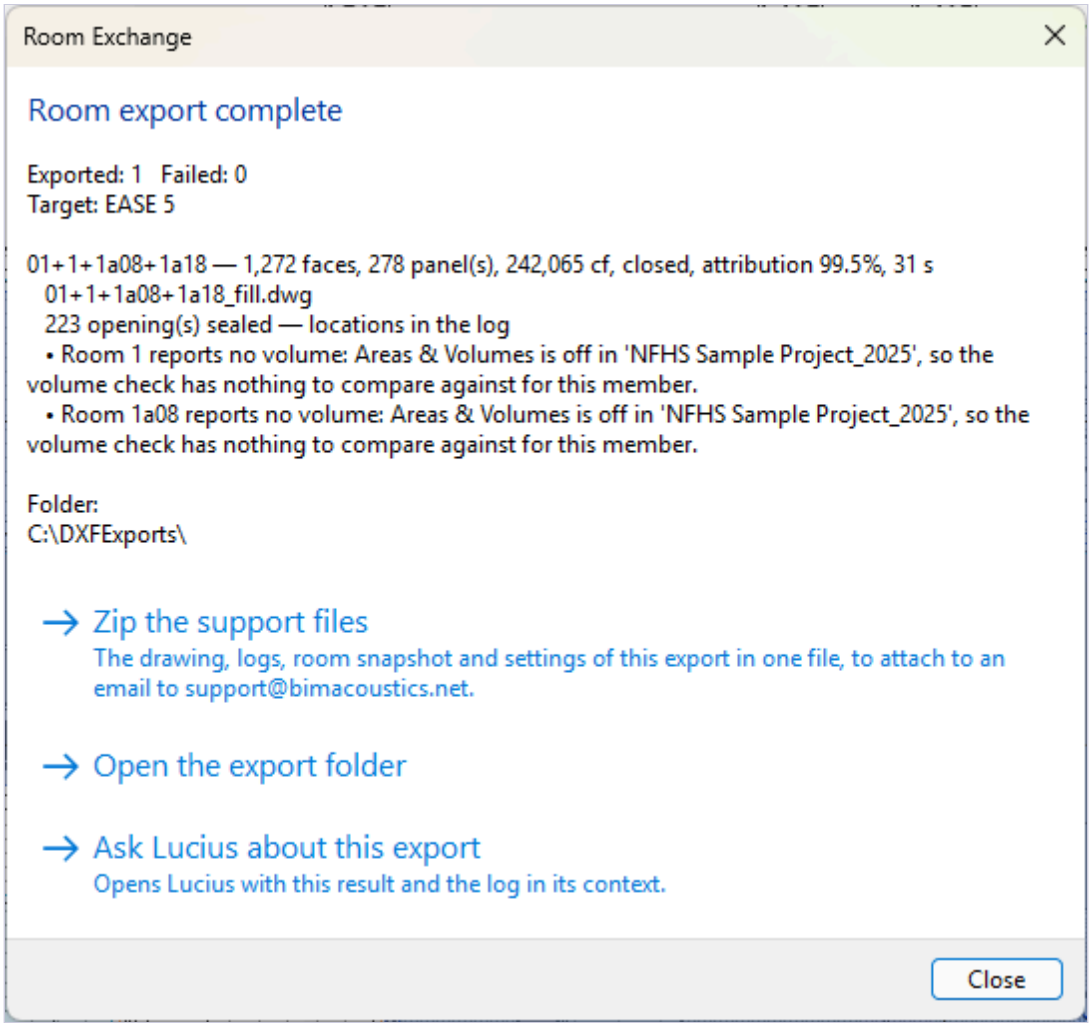


Figure 12. The result: 1,272 faces, closed, 99.5 % attribution, 242,065 cf, 31 seconds.

In EASE the volume is green and the 3D view shows the house, the raked seating, the stage, the catwalk and the pit as one room.

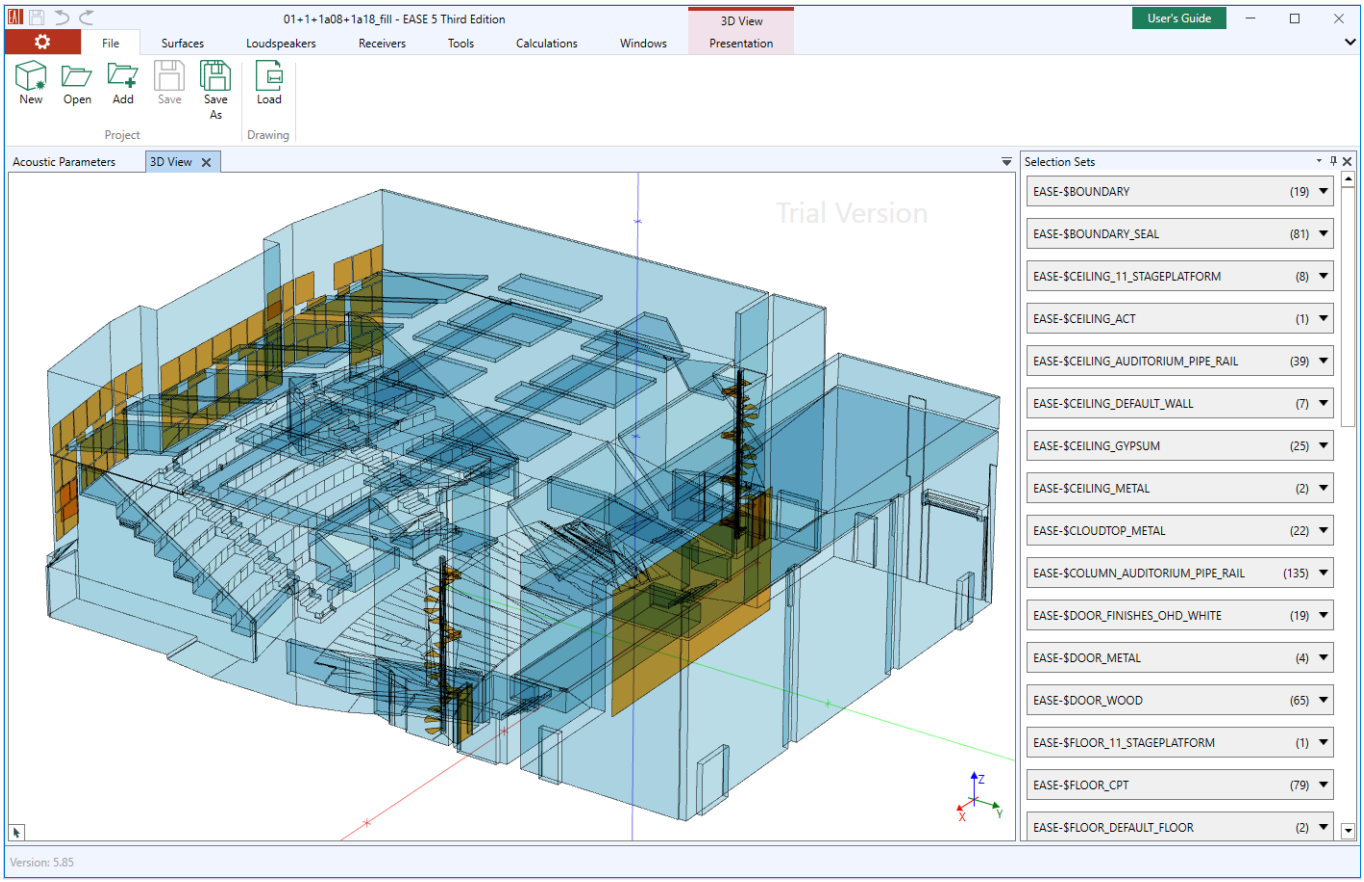


Figure 13. The theater in EASE 5: one closed volume from the pit to the roof.

How to tell the rooms did not cover the space

- The result dialog says **the air reaches the limit of the ticked rooms toward room X over N sf with no surface there**. Either tick room X, or the ticked rooms stop short of a surface there.
- EASE shows a large flat face on `EASE-$BOUNDARY` where you expect a ceiling, a floor or a wall.
- The volume in the result dialog is well short of the room's real volume.
- A balcony, a catwalk or a stage is missing from the EASE 3D view.

In each case, fix the room in Revit, not the drawing in EASE.

2. The origin marker

EASE needs to know where 0,0,0 is and which way the room faces. Room Exchange takes both from an **origin marker**: the family JSBA Room Ref Origin. The first time Room Exchange runs in a project it loads the family for you and says so; you only place one instance per venue, with the Component tool like any other family.

Where to place it. Inside the room, on the floor, at the point you want as EASE's origin. For a theater that is the centre of the stage front at stage level; for a flat room, the centre of the front wall at floor level. The marker must stand inside the room's air, so not inside a wall; a marker on the curtain-wall sill is fine, one in the corridor outside is not.

Which way to turn it. The marker's own axes become EASE's axes: its +Y points upstage, toward the stage from the house; its +X points house right; Z is up. Rotate the marker to suit the room. Never mirror it: a mirrored marker gives a left-handed frame and every loudspeaker aim comes back reversed.

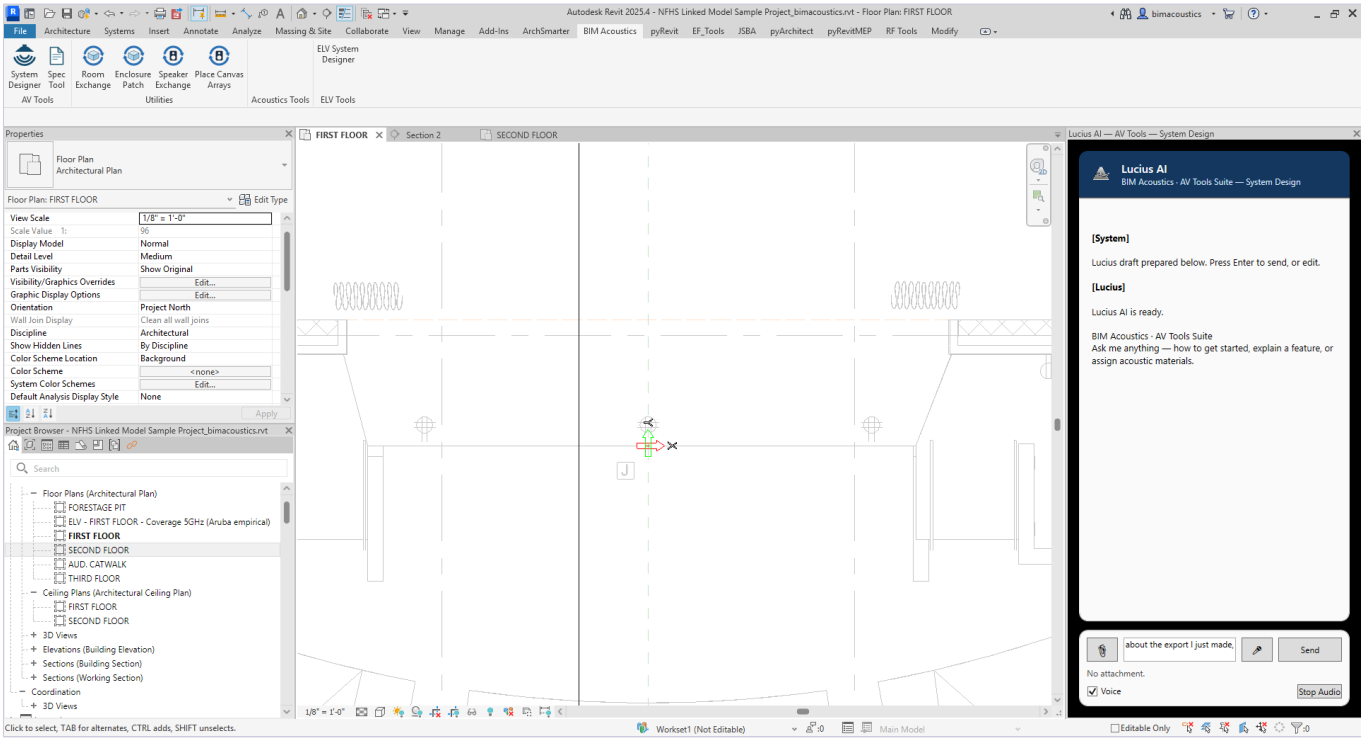


Figure 14. The origin marker placed at the centre of the stage front on the first floor plan. Its green axis (+Y) points upstage, its red axis (+X) house right. On the right, the Lucius pane opened from the result dialog with the question prefilled.

One marker per venue. Room Exchange uses the marker inside the ticked rooms. If none stands inside them it takes the nearest one and says so in the result dialog: *the origin marker used is more than 1 ft outside the captured rooms*. Two rooms exported separately can share one marker if it stands in one of them and you accept the note for the other; a venue with its own origin wants its own marker.

If a room has no marker inside it, Room Exchange asks before it starts, naming the rooms: *Place a marker now* (click the point in a plan view, press the space bar to turn it, press Esc twice to carry on; place one in each room listed), *Continue anyway*, or Cancel. A marker standing in another room does not count: each export takes the marker inside its own rooms, and rooms exported together as one need one marker among them. *Don't ask again* turns the question off (`SkipOriginMarkerPrompt` in the settings file, section 8).

Turn the marker square to the room. Room Exchange lays a room out along its walls before it fills it. When the walls alone give no clear direction (a fan-shaped hall, a room with several wall directions), the wall direction nearest the marker's red arrow (+X) is used, so walls that run along the marker come out straight instead of stepped. A marker a few degrees off is fine; one nowhere near any wall is ignored.

Without a marker the export is written in Revit's internal coordinates, the result dialog warns *no origin marker in the project*, and the drawing may land hundreds of feet from EASE's origin. It still imports, but loudspeaker positions brought back through the Import Loudspeakers tab have no shared frame to come back by, so place the marker before the export you intend to work from.

The marker's origin and axes are written beside every drawing in a small origin file (`<name>.frame.json`). The Import Loudspeakers tab reads it to put EASE loudspeakers back into the model at the right place and aim. The settings key `RefMarkerFamilyNameContains` names the family Room Exchange looks for; change it only if you use a marker family of your own.

3. The window

The window floats above Revit: while it is open you can still pan, zoom, select and look around the model (for example to see which rooms to tick, or to place a host room). Nothing happens to the model until you press a button (Export, Capture Snapshot Only, Custom Perimeter or Place Loudspeakers), and the window closes when you do. The window reads the model when it opens; if you change the model while it is open (add a room, load a loudspeaker family, place an origin marker), press **Refresh** on the Rooms tab or the Import Loudspeakers tab. It reads the rooms, the loaded families and the origin markers again; the rooms you had ticked stay ticked, and your family picks and Place ticks on the Import tab stay where they are still on offer. If a floating window gives you trouble, set `KeepWindowModal` to `true` in the settings file (section 8) to get the older window that holds Revit until it is closed.

Rooms tab

The list shows the rooms of the model and of its linked models, with number, name, area, level and source. Find narrows the list; Show ticked only hides the rest; Tick Listed ticks everything the filter shows.

- **Export** exports the ticked rooms. Rooms that touch go out as one file after the merge prompt; rooms with no connection go out as separate files.
- **While the EASE model is built** a small window shows the time so far and a **Cancel** button. Room Exchange reads the model in Revit first, then builds the EASE model in a separate program beside the add-in, which is many times faster than building it inside Revit; Revit waits until it is done. **Cancel** stops the export, and nothing is written. If the separate program cannot run, the model is built inside Revit as before and the result says so (section 5).
- **File name** is optional. Empty names the files after the room numbers with the suffix `_fill`. A typed name is used as typed, cleaned of characters a file name cannot hold. When one export writes several files, the room numbers are appended.
- **Export for**: DWG (EASE 5, other) is the normal choice for EASE 5; it needs AutoCAD on the machine, which Room Exchange uses only to convert the DXF it writes to DWG. DXF (Bose Modeler, EASE 4, other) is for Bose Modeler, which reads DXF only, and for EASE 4. Room Exchange is supported with EASE 5. The exported DXF and DWG files also appear to import into Bose Modeler and ODEON, but this has not been fully tested and is not supported; they may work in other programs that import DXF or DWG files containing 3D polylines and/or 3D faces.
- **Custom Perimeter…** traces a boundary by clicking in a plan view, for a space no placed room describes, or whose room cannot be used (for example a linked room you cannot change, or a room Revit cannot build the volume of). You give it a number, a name, a base (level plus offset) and a top (level plus offset). The perimeter is **saved in the project** as a Filled Region, with a see-through Generic Model volume between its base and top so you can see it in sections and 3D, and it is **listed in the**

room list (Source: Perimeter) from then on, like a room. Tick it and Export, or use **Edit Perimeter...** (click its row first) to change its number, name, base or top. It goes through the same engine as a room: Room Exchange fills the space inside the outline and takes the real walls, floors and ceilings it finds there, in this model and in its links. Rooms inside a perimeter do not cut it up. **Adjust in 3D...** (click the perimeter's row first) opens a 3D view with a section box on the perimeter's base and top, and the perimeter's volume drawn in it. Drag the box's bottom and top against the real floor, roof and ceilings, then press **Save heights** in the small panel (the perimeter and its volume take the box's heights and the panel stays open) or **Export** (it saves them and exports). **Trace & Adjust in 3D** in the Custom Perimeter dialog does the same right after tracing. Closing the panel keeps the saved perimeter. A perimeter lives in the Revit file: **save the model** (Ctrl+S) after tracing one, or it is gone when Revit closes without saving. A boundary Filled Region drawn before heights were kept, or by the System Designer's temporary-room tool, is listed as **Perimeter (set height)**: its base is the level of the view it was drawn in and its top a default (the stamp's ceiling height when it has one); use Edit Perimeter to set the real heights. The volume is a drafting aid and is never part of the export; hide it with the Generic Models category in a view if it is in the way. The older engine is still available behind `UsePerimeterCarve` in the settings file.

A perimeter takes in every space inside its outline that the air reaches: rooms inside it do not stop the air, and an open doorway, a cased opening or a gap into a back-of-house room lets the air into that room too. Unlike ticked rooms, there are no openings at room edges inside a perimeter. Trace along the space's own edge, not the outside walls of the building, to keep such rooms out.

One perimeter per level

When the outside walls of a space change from level to level (an atrium whose edge steps back on each floor), trace one perimeter on each level and export them together:

1. Open the plan of the lowest level and press **Custom Perimeter...**. The base starts at that plan's level and the top at the next level up; check the top, because the next level up can be a mezzanine level rather than the floor above.
2. Press **Trace & Save** and trace the outline (Esc when finished). The perimeter is saved without exporting, and the room list comes back with it ticked, together with what was ticked before.
3. Open the next level's plan and repeat, until every level has its perimeter.
4. Press **Export**. With two or more perimeters ticked, Room Exchange asks **Merge Perimeters**: **Yes** (the default) exports them as one model, **No** exports each on its own.

In the merged model a perimeter stops at the floor of the perimeter above it, where that floor is solid; where the floor is open (the atrium void, a stair opening) the air runs on into the perimeter above. Ticked rooms are exported apart from the perimeters.

Curved outlines

Trace joins the points you click with straight lines; it has no arc. For a space bounded by curved walls, trace a rough outline first and then correct it with Revit's own sketch tools, which have arcs and Pick Lines:

1. **Custom Perimeter...**: give the number, name, base and top, press **Trace & Save**, and click a rough outline in the plan (Esc when finished). The perimeter is saved without exporting, and the room list comes back with it ticked.
2. In the same plan view, select the perimeter's Filled Region and click **Edit Boundary**.
3. Replace the straight stretches along the curved walls: **Pick Lines** on the walls themselves, or draw them with an arc tool. If Revit will not let you pick a wall of a linked model, draw the arc instead. Keep the outline one closed loop: only the outer loop is read, and a loop drawn inside it is ignored.
4. **Finish Edit Mode**, and save the model.

Room Exchange reads the edited outline at the next export; arcs go out as short straight lines. The base and top stay as they were saved. The see-through volume keeps the old outline until you open **Edit Perimeter...** on the perimeter and press **Save**.

Options tab

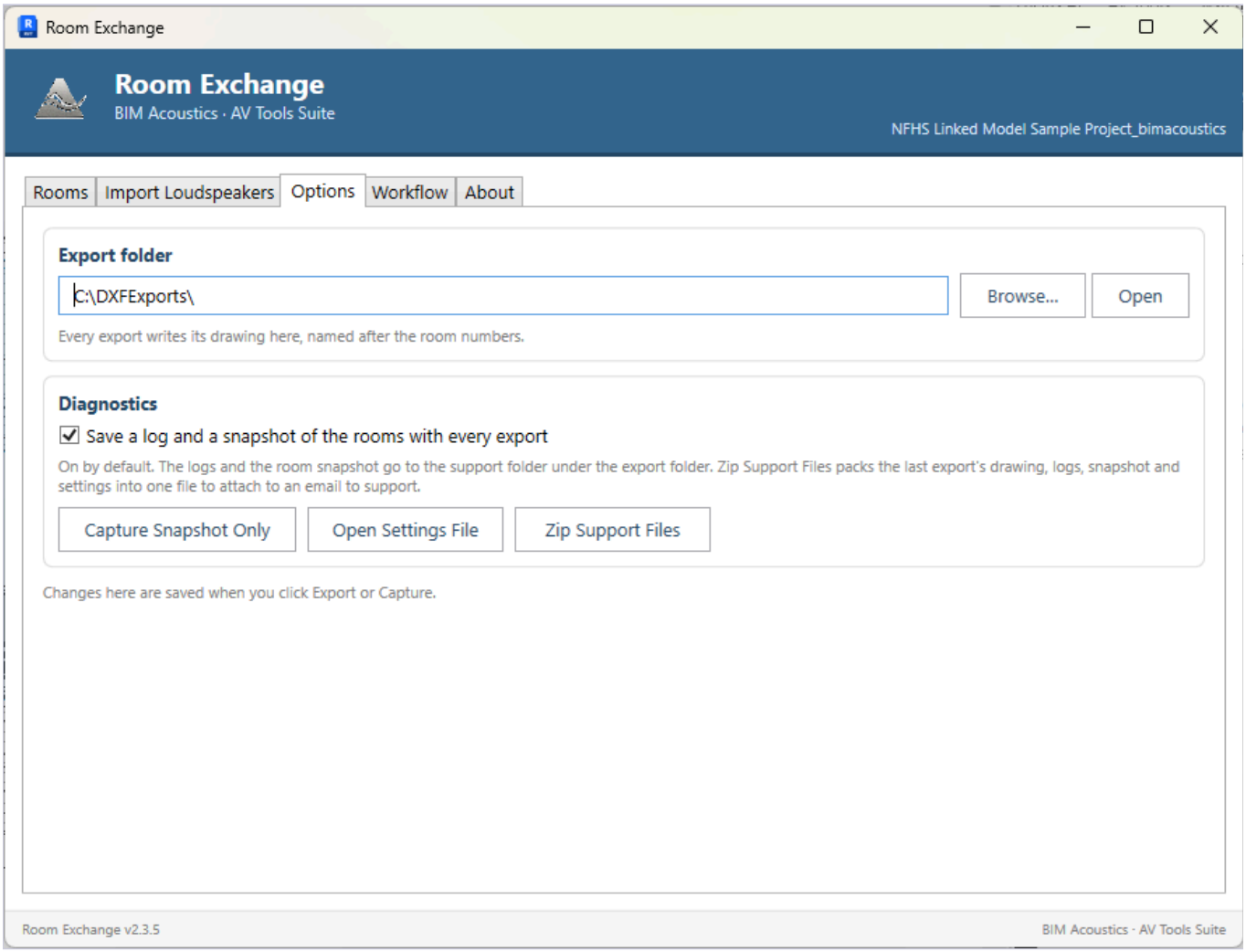


Figure 15. The Options tab: export folder, Diagnostics on, Capture Snapshot Only, Open Settings File and Zip Support Files.

- **Export folder**: where the drawings go, by default `C:\DXFExports\`. The `support` folder under it holds the logs and the room snapshots. The Rooms tab shows the same folder in an **Export to** line above the Export button, with **Browse…** and **Open**, so you can see where an export will go and change it without leaving the tab. The folder is remembered for every project.
- **Diagnostics**: on by default. Writes a capture log, a fill log and a room snapshot for every export. Leave it on; the snapshot is what lets support re-run your export without your model.
- **Curved surfaces → Panel size (ft)**: a curved ceiling or roof is written as flat panels of about this size, instead of one small face for every triangle of the Revit model. The panels run along the element's own direction; every fold sharper than 5° stays a hard edge, and a strip narrower than the minimum feature size (3.5 in) joins the surface beside it. Each element gets at least six panels across its shorter side, and no panel is smaller than 1 ft. Only the **Ceilings** and **Roofs** categories are changed; every other category is written as before. An element is left as it is when panels would not reduce its number of faces or would open a gap in the room's surface; the fill log says which elements were changed and why others were not. Saved with each project; the default is 4 ft, and 0 turns it off.
- **Capture Snapshot Only** saves the snapshot of the ticked rooms without exporting.
- **Open Settings File** opens your settings file (section 8).
- **Zip Support Files** packs the last export's drawing, origin file, linework notes, logs, snapshot, the settings file in use and the remembered checklist choices of the model into one zip in the support folder, and opens the folder with it selected. The zip is named `RoomExchange_support_<date>_<time>.zip`: the model's name and the room numbers are not in the file name. They are written inside, in `versions.txt`.

Workflow and About tabs

The Workflow tab is the short version of this guide. The About tab shows versions and file locations, and has Zip Support Files and Email support.

Import Loudspeakers tab

This tab brings loudspeakers designed in EASE back into the Revit model. It reads the loudspeaker list (the `.xld` file) saved from the EASE project that was built on a Room Exchange drawing, and the origin file written beside that drawing (section 2), so each loudspeaker lands at its position in the room.

- **EASE loudspeakers**: the `.xld` file. **Family folder**: the folder of loudspeaker acoustic data, used to find the Revit family for each EASE model. **Origin**: the origin marker the drawing was written from; **Origin File…** picks the origin file instead when the marker is not the one in the project.
- **The list** shows each loudspeaker's label, EASE model, position, pitch, azimuth and roll, and the Revit family it matched. The **Match** column says how: *contains* or *chosen* means a family was found; *2 fit, choose one* means several families fit, so pick one from the list; *no family found* means none was found. Room Exchange also looks in the family folder: a family that comes with a type catalog (a `.txt` file beside the `.rfa`) is matched by its types, and when the whole name does not match, the distinctive word of the model is used (for example MKC80 in "MKC80 EAWF").
- **Long family lists.** Each loudspeaker's family list shows the suggested families first, then a gap, then every other family. **Filter families** above the table shortens all the lists to the names that hold the

words you type (for example `mkc`, or `eaw black`); **Clear Filter** shows them all again. The filter only changes what the lists show: the family chosen for a loudspeaker always stays in its list. Choosing the folder of one manufacturer as the family folder also keeps the lists short.

- **Pick once.** When you pick a family for one loudspeaker, the other rows of the same EASE model that have none take it too, and it is remembered for the next import.
- **Families without a type catalog** show their types only once they are loaded in the project. Load the family into the project (Revit: Insert, Load Family) and the loudspeaker is matched to its types; or pick the family by hand from the list.
- **A loudspeaker with no family is not placed.** Pick a family that is already loaded in the project and is the right one, or load the correct family into the project and run the import again.
- **Place** is ticked for every row. Untick a row to leave it out. **Place Loudspeakers** places all the ticked rows, normally in well under a second, and a result lists how many were placed, how many were aimed again and how many were not placed, with the reason for each.
- **Already in the model.** If a loudspeaker of the chosen family already stands at an EASE position (within an inch), you are asked first: *Yes* aims those again with the EASE angles and places the rest, *No* leaves them as they are and places only the others, *Cancel* goes back to the list. No loudspeaker is placed twice.
- **Aim.** EASE's pitch is the vertical angle (negative is down) and its azimuth is the horizontal angle (positive is counter-clockwise from above). The front of a Revit loudspeaker family faces the family's -Y, which is 180 degrees from EASE's zero, so Room Exchange applies a 180 degree front offset by default. **Parameter Mapping...** shows how each family takes the angles and lets you change it; a mapping saved for a family is used instead of the default. The azimuth is applied by turning the loudspeaker instance in plan, so any family works whatever its own azimuth parameter does; name an azimuth parameter in Parameter Mapping only if the family needs one. A pitch or roll box left blank is found automatically when the loudspeaker is placed, and "(not written)" leaves that angle out.

4. The checklist

Between Export and the capture, Room Exchange shows **Elements in the Export**, the list of everything inside the ticked rooms that is not plain building envelope. This is where you decide what the acoustic model treats as present and what as absent. Walls, floors, ceilings, roofs, columns and structure are the envelope; they are never listed and are always in.

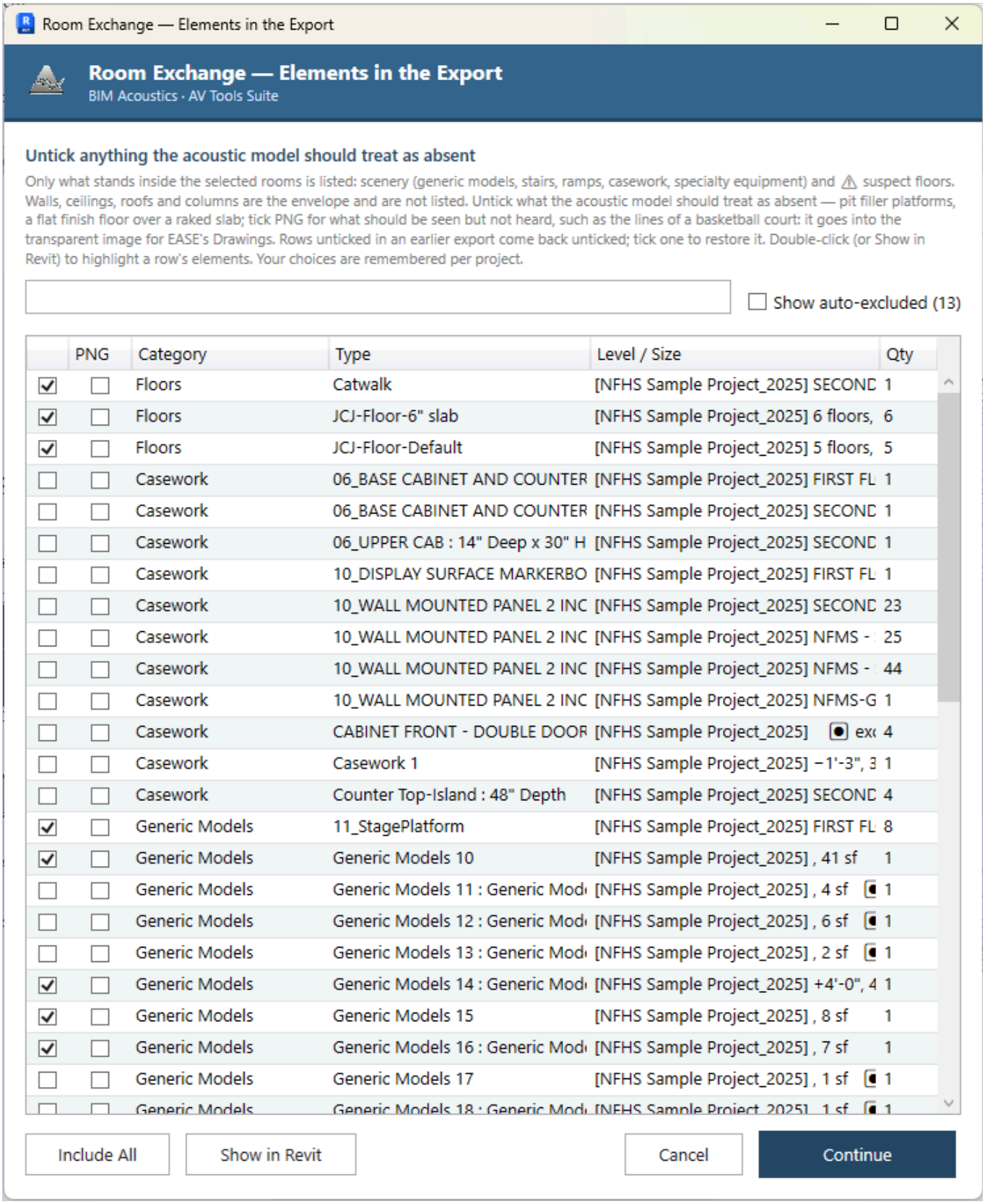


Figure 16. The checklist for the theater. Floors, the stage platform and the generic models that are structure stay ticked; casework is left out; 13 auto-excluded rows are hidden.

What is listed

- **Generic models, stairs, ramps, casework and specialty equipment** standing inside the ticked rooms, one row per element. Elements of one type that occur three or more times share one row, with the count in the Qty column; one tick decides for all of them.
- **Suspect floors**, marked ⚠: two substantial floors overlapping in plan at different heights. That is the pattern of a flat finish plane drawn over a raked slab, or a filler over a pit. Ordinary floors are not listed.
- **Rows from linked models** carry the link's name in square brackets at the start of the Level / Size column.
- **Casework and furniture** inside the rooms. These are listed so you can see them, but they are not exported.

The columns

- **First column, the tick**: ticked means the element is in the acoustic model and blocks air like a wall; unticked means the model treats it as absent. Scenery, a pit filler platform that is normally out, a flat finish floor drawn over a raked slab, a temporary stage: untick them.
- **PNG**: seen but not heard. The element's lines go into the transparent plan image for EASE's Drawings and the element is not in the acoustic model. Use it for court lines, the roof steel you want to see when placing loudspeakers, or anything that helps you orient in EASE. Ticking PNG clears the first tick.
- **Category, Type**: the Revit category and the family and type name. A bare type name is often a model number, so the family name is shown in front of it.
- **Level / Size**: the level and offset for a single element, the span for a group, and a note on why the row is there: the ⚠ pair it belongs to, ⛔ for an automatic exclusion, ● for a choice remembered from an earlier export.
- **Qty**: how many elements the row stands for.

Automatic exclusions

Two kinds of floor are unticked without asking, because they are never structure and ticking them lids the real floor at the wrong height: thin finish films, a floor whose whole thickness is under about 2 inches, and floors named "rendering only". They are marked ⛔ and hidden. **Show auto-excluded (N)** lists them; tick one to bring it back.

Demolished elements

Elements that are not part of the finished building are left out of the export whatever the list says: anything with a **Phase Demolished** set, in this model or in a link, and anything created in a phase whose name contains "Demo". This matters most for a renovation, where the existing building's model still holds the floors, ceilings and walls that the new design removes: left in, an existing floor can close a space off at the wrong height. The result dialog gives the count for each model (section 5). Walls are never listed in the checklist and floors only when suspect, so for them this is often the only way out of the model. If a demolished element should be in the acoustic model after all, clear its Phase Demolished in the model it belongs to.

Working the list

- Type in the filter box to narrow the list by category, type or level.
- Select several rows with Shift or Ctrl, then tick one: the tick applies to the whole selection.
- **Show in Revit**, or a double-click, highlights the row's elements in the model, or zooms to them when they are in a link.
- **Include All** ticks every visible row.
- **Continue** runs the export with the list as shown. **Cancel** stops the export; nothing is remembered.

Choices are remembered

Your ticks are stored per project. A row you unticked comes back unticked in every later export of rooms that contain it; tick it to restore it. Rows for elements in other rooms are left as they were. The stores live under your user profile and are included in the support zip.

What to leave ticked

Stage platforms and their risers, stairs that stand in the room, catwalk floors, stage lifts and pit fillers in the position the room is normally used in, display cases and counters that are built in. When in doubt, leave it ticked and look at the result in EASE; a wrong inclusion shows as an odd box, a wrong exclusion as a missing surface.

5. The result dialog

Each exported volume gets one block of lines.

- **faces, panel(s), cf, closed or open edge(s), attribution %, s**: the face count, the two-sided panels (clouds, pads, curtains, bleacher boards), the volume, whether the shell is closed, how much of the surface is named from a real element, and the time. A volume that says **closed** imports green.
- **N opening(s) sealed**: gaps in the model under about 2.5 ft that the fill closed. Normal; the seals are on `…_BOUNDARY_SEAL` layers.
- ► **The air reaches the limit of the ticked rooms toward room X over N sf with no surface there**: the ticked rooms open into room X with nothing between. Tick X if the spaces are one, or find what should close the opening. 50 sf is a doorway; 5,000 sf is a missing ceiling or a shared wall.
- ► **N sf of opening to room X across a room separation line**: tick X if it is part of the space; otherwise the opening is on the `EASE-$OPENING` layer for you to give an opening material.
- **Room X reports no volume: Areas & Volumes is off**: the model does not compute room volumes, so the exported volume cannot be checked against Revit. Harmless.
- **Room X stops N ft below the deck: the region was raised to the deck**: the room's upper limit was below the real ceiling or deck, so the fill was allowed to rise. If the room truly stops lower, a ceiling in the model will stop it.
- **Room X's export stops at N ft above the floor, but the space continues up to about M ft**: the room's own height (its Upper Limit) is lower than the real space, and what the export used as its top has nothing above it. A large part of the room's top (10 percent of its plan or more) is a flat lid on the

`EASE-$BOUNDARY` layer instead of a real ceiling or roof. Room Exchange looks straight up from the lid in the model and its links for the first ceiling, roof or floor and names the height it found. Raise the room's Upper Limit (and Limit Offset) in Revit, or use the link **Extend room X … and export again** under the result: it exports once more with the room's top lifted to the ceilings and roofs found above, and leaves the room in Revit as it is. If the same message comes back, the model above the room is not what Room Exchange looks for (for example a roof built as a generic model), or the space is open to the sky by design.

- **Perimeter X: the export stops at N ft above the base…**: the same for a Custom Perimeter whose top is below the real ceiling or roof. Raise its top with **Edit Perimeter…**, or use the link **Extend perimeter X … and export again**, which lifts the top for that export only and leaves the saved perimeter as it is. A top you will use again is better saved with Edit Perimeter.
- **N demolished element(s) (a Phase Demolished set, or created in a Demo phase) were left out: … in <model>, …**: how many elements of each model were left out as demolished (section 4).
- **The fill ran inside Revit, which is much slower: <reason>**: the separate program that builds the EASE model could not be used (it is not installed, or the room snapshot could not be saved), so the model was built inside Revit. The result is the same; only slower.
- **The origin marker used is more than 1 ft outside the captured rooms**: the drawing's origin comes from the nearest Room Ref Origin marker, which stands in another room. Place one in this room for a clean origin.

The links at the bottom zip the support files and open the export folder. The dialog holds a limited number of links, so when the connected-rooms link below also appears, the link that opens the export folder can be left out (the folder is named in the text above it). When the space is open into rooms that are not ticked, a fourth link, **Add the N connected rooms and choose again**, opens the room list with the last selection and those rooms ticked. It offers only rooms the air actually reached: a room behind a door is never offered, because a modelled door stops the air; a room appears when nothing stands between (a cased opening, an arch, a room separation line, a missing wall). Left off, with the reason in the fill log: outdoor and service spaces (terraces, mechanical rooms), the same space drawn again in another link, and envelope rooms that span half the building. Change the ticks as you like and press Export. The same link also unticks any ticked room that came out as its own box, not joined to the main volume, and lists it; keep one ticked if it is meant to be separate. Each round can reveal the next ring of rooms, so repeat until the result stops offering any. The rooms it adds are suggestions from the fill: check the list before you export.

A slow export: when the capture of the rooms took more than a minute and a half, mostly for one or two heavy elements (a Direct Shape of hundreds of thousands of triangles, often a piece of equipment or a model of the building skin that stands outside the rooms), a window opens after the result dialog. It names each element, says whether it is inside or outside the rooms, and has a **Show in Revit** button so you can see what it is. **Leave out and export again** adds it to this model's exclusion list and exports once more, which is much faster; **Keep it** leaves everything as it was. An element outside the rooms does not change their surfaces; one inside a room stays in the checklist, unticked, so you can put it back.

6. Reading the import in EASE 5

Load the drawing with File → Load. On the Acoustic Parameters page the Volume tile has a colour bar:

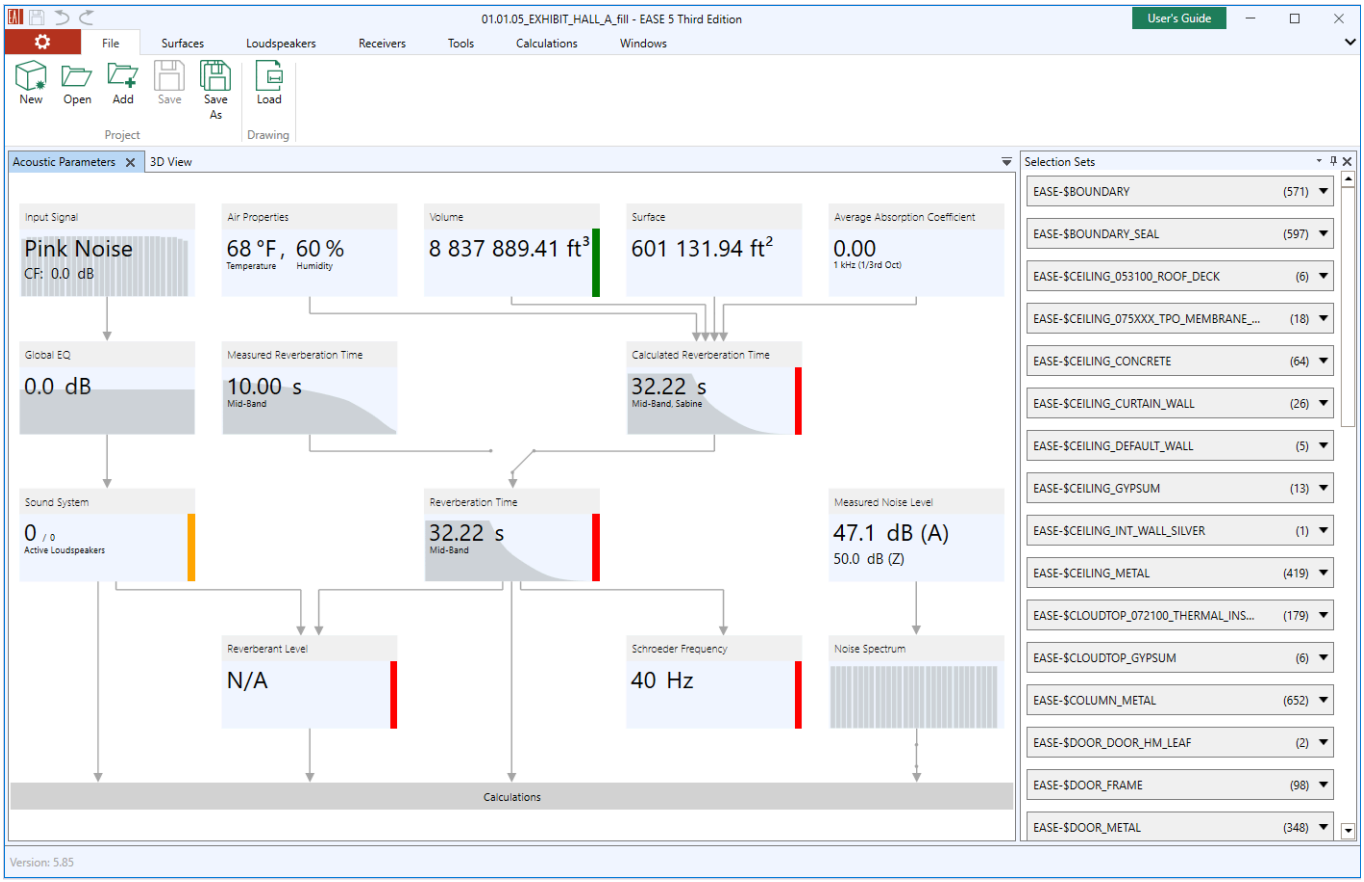


Figure 17. Exhibit Hall A on the Acoustic Parameters page: the green bar beside the volume means the room is closed. Compare the volume with the result dialog's.

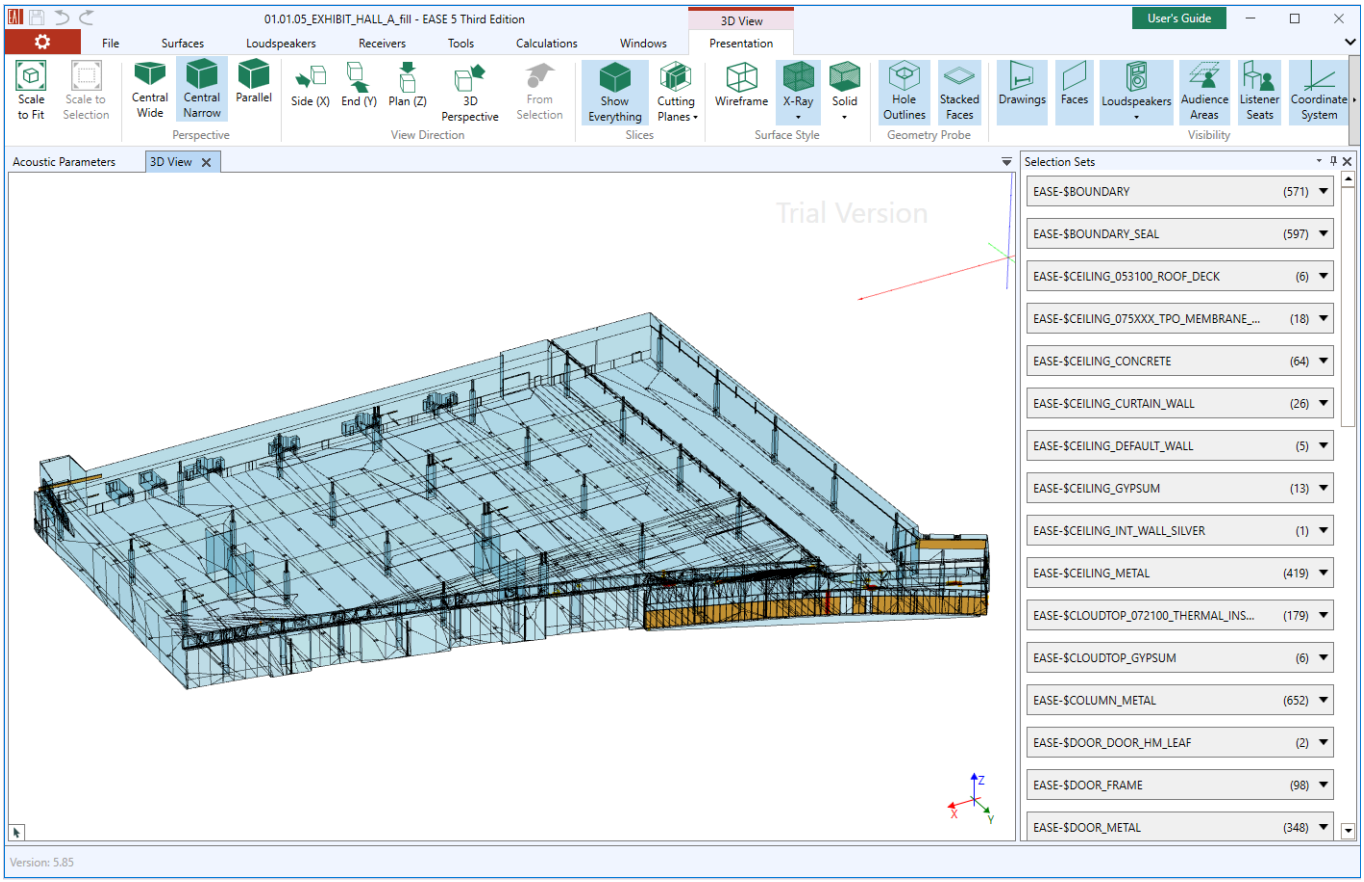


Figure 18. The same hall in the 3D view: a 220,000 sf exhibition floor with its roof structure, truss walkways and show-power trenches, exported from one linked room. A big room takes minutes to export, not seconds.

- **Green**: closed. Use it.
- **Yellow**: small open edges only. Hole Outlines in the 3D view shows them. Hairline seams where a floor meets many wall bases are harmless. A long straight yellow line across a face means a face was written twice or is missing; send the support zip.
- **Red**: not closed. Hole Outlines shows where. The usual cause is a room that opens into an unticked neighbour; the result dialog's ► lines name it.

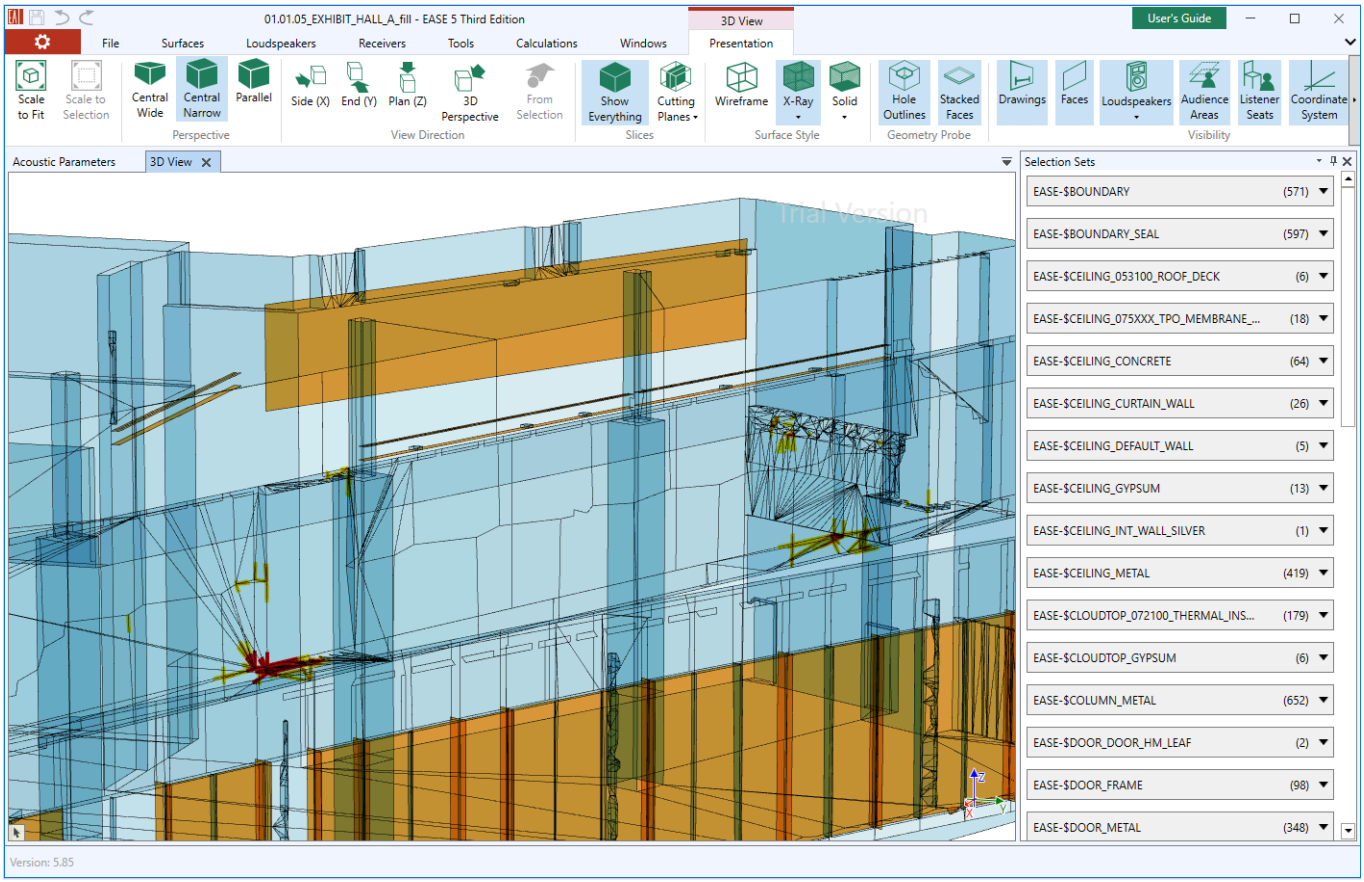


Figure 19. Hole Outlines on, in the same hall. The volume is green; the few yellow and red outlines on a narrow angled ledge are stacked faces from a sloped soffit, a known limit, and do not open the room.

If the Import Log tab is not there, EASE refused nothing: yellow or red is geometry, not a rejected face. EASE's volume should be within about 1 % of the result dialog's.

7. Layer names

Every layer is an EASE Selection Set.

The layer name is not the EASE material. Room Exchange names each layer after the Revit material or family it found on the room side of the surface, so you can tell the surfaces apart. Which acoustic material each layer gets in EASE is your decision when you assign materials there; Room Exchange assigns nothing. A layer called WALL_GYPSUM tells you those faces are gypsum board in the model; you choose the gypsum-on-studs data set that fits.

Layer	What it is
`EASE-$WALL_<material>`, `EASE-$FLOOR_<material>`, `EASE-$CEILING_<material>`	A surface of the room, named by the material on the room side, or by the family name for a loadable family.
`EASE-$DOOR_<material>`, `EASE-$DOOR_FRAME`, `EASE-$WINDOW`, `EASE-$STAIR_<type>`, `EASE-$COLUMN_<material>`	The same for those elements.
`EASE-$WALL_CURTAIN_WALL`, `EASE-$CEILING_CURTAIN_WALL`	Curtain wall panels and mullions.
`EASE-$WALL_OPERABLE_PARTITION`, `EASE-$CEILING_OPERABLE_PARTITION`	An operable partition, its track, and any thin wall or glazed leaf on its line.
`EASE-$CLOUD_<material>`, `EASE-$CLOUDTOP_<material>`	A floating ceiling or cloud: CLOUD is its underside facing the room, CLOUDTOP its back facing the plenum. Both carry the cloud's own Revit material in the name. In EASE give CLOUDTOP the material of whatever lies on the back of the cloud: the insulation blanket if there is one, otherwise the cloud's bare back (gypsum, metal pan).
`EASE-$RAILING_<material>`	The panels of a railing (a glass guard round a stair opening or a mezzanine edge), each a closed slab of the pane's own thickness, cut where it enters a stair or a wall. Rails, posts and balusters are left out as detail.
`EASE-$FRAMING_<material>`	Structural framing in the room's air: each member one two-sided panel along its web, the member's depth tall, cut where it leaves the room. Untick a framing type in the checklist to leave it out, or tick its PNG box to see it in the plan image only.
`EASE-$PANEL_<family>`, `EASE-$WALL_<family>` with `EASE-$PADBACK_AIRGAP`	Wall pads, acoustic panels and curtains drawn flat; the rear face is the air gap behind a pad.
`EASE-$BOUNDARY`	The edge of the ticked rooms where no surface stands. See section 1.
`EASE-$BOUNDARY_SEAL`	A small gap the exporter closed.
`EASE-$OPENING`	An opening across a room separation line.

Faces under 1 sf take the layer of the larger face beside them, and a material that appears mostly as walls with a few square feet as floor or ceiling is folded into its wall layer, so the list stays short.

A thin element written as a two-sided panel (a screen, a glazed panel, a baffle) stops where the room's air ends. A curved one that runs out of the room, such as a screen wall curving along a whole floor, is written as flat pieces along its curve, each cut where it leaves the room.

8. Settings

`RoomExchangeSettings.json`, from Options → Open Settings File. Only these matter to a user.

An installed copy keeps its settings file in the program folder, which a standard user cannot change. The changes you make, such as the export folder on the Options tab, are saved to a copy in your own folder (`%AppData%\JSBA\RoomExchange\RoomExchangeSettings.user.json`), and Open Settings File opens that copy. The newer of the two files is the one Room Exchange reads.

Key	Meaning
`ExportLocation`	The export folder.
`DebugMode`	1 = Diagnostics on (default).
`SkipLinkTitles`	Linked models, or categories within them, to leave out. `"FURN"` leaves out the whole furniture model; `"FURN: Walls"` leaves out only its walls. Comma-separated.
`KeepWindowModal`	`false` (the default) lets the Room Exchange window float so Revit stays usable while it is open. `true` brings back the older window that holds Revit until it is closed.
`SkipOriginMarkerPrompt`	`false` (the default) asks to place an origin marker before an export when a room has none inside it. `true` stops the question.
`UsePerimeterCarve`	`false` (the default) exports Custom Perimeter through the fill engine. `true` brings back the older carve engine for it.
`PlanarSlackFt`	`0` (the default) keeps the writer's built-in 0.02 ft (a quarter inch): a polygon whose corners lie farther than that off its plane is split into triangles. A larger value, for example `0.05` or `0.1`, snaps more polygons flat and writes far fewer faces on curved ceilings, at the price of cracks of that size between neighbouring faces. A test setting: check in EASE that the model still reads as closed.
`OperablePartitionKeywords`	Words in a family or type name that mark an operable partition. Built in: operable, hufcor, kwik-wall, modernfold, skyfold, moderco, folding partition, movable wall, movable partition.
`FoldedSheetKeywords`	Words that mark a curtain, drape or pleated panel drawn flat as one sheet. Built in: pleat, curtain, drape.
`RefMarkerFamilyNameContains`	The origin marker family, "Room Ref Origin".

9. Known limits

- Open-air spaces (stadium bowls, terraces) get a synthetic lid at the room's upper limit. There is no open-room mode yet, and EASE 5 has no open-room calculation.
- Very large spaces take a while: a four-level atrium of about 10.7 million cf took about 15 minutes, some 10 of them reading the model in Revit and 5 building the EASE model; a ballroom takes a few minutes,

a meeting room under a minute. Reading the model is slowest where it holds heavy detail (structural steel, escalators, detailed ceilings); unticking what the acoustic model does not need shortens it.
- A very large export can make a model with hundreds of thousands of faces. Each ticked element adds faces: steel framing, escalators and detailed ceilings add the most.
- Furniture and seating are not exported.
- A perimeter's volume and outline are edited with Edit Perimeter, not by moving the Revit elements: a moved Filled Region changes the outline but not the volume until the perimeter is edited and saved again.
- A linked room whose boundary cannot be read cleanly gets a warning; place a host room or use Custom Perimeter.
- Narrow angled ledges can leave a few stacked faces that EASE outlines; harmless.

10. Sending a problem

1. Options → Zip Support Files, or the link on the result dialog. The zip is in the `support` folder under the export folder, named `RoomExchange_support_<date>_<time>.zip`.
2. Email the zip to support@bimacoustics.net with a sentence on what looked wrong, and a screenshot of the EASE 3D view with Hole Outlines on if you can.
3. The zip holds the drawing, logs, the room snapshot, the settings file in use and the remembered checklist choices of the model you exported (not of your other projects). The snapshot contains only the rooms you exported and what stands around them. The model's name and the room numbers appear inside the zip, never in its file name.